Auburn Governance Report

Verification Layer Architecture for
AI-Assisted Formal Mathematics

A Reference Standard for the Auditing and Governance of
Machine-Verified Mathematical Results Produced Under
Structured Human-AI Collaboration

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Purpose. This document defines the complete verification layer architecture required for governance-grade confidence in AI-assisted formal mathematics. Each layer is independently load-bearing: it closes a class of failure that no other layer addresses. The architecture is designed so that no single point of failure — human, machine, or procedural — can produce an undetected error in the final result. This is not a description of what was done for one paper. It is a reference standard for what must be done for any result of comparable stakes produced under comparable conditions.

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1 Overview

The verification of a mathematical result produced through AI-assisted collaboration requires more than correctness of the mathematics. It requires governance: a structured system of independent, overlapping verification layers, each closing a specific class of failure that the others cannot address.

The architecture consists of eleven layers, numbered 0 through 10. Each layer has a defined scope, a defined failure class it addresses, and a defined independence property that prevents it from inheriting the blind spots of adjacent layers.

Layer 10: Living Failure Mode Standard Failure mode taxonomy created, audited, and unified across systems as the audit proceeds.
Layer 9: Per-File Coq Review Under Both Protocols One file at a time, under FIELDS and FIELDS-COQ simultaneously.
Layer 8: Sub-Section-Level Review Under FIELDS Protocol Every numbered sub-section and result individually evaluated.
Layer 7: Block-Level PDF Review Across Tier-1 Systems Section-sized blocks reviewed independently across each system.
Layer 6: Independent Verification Infrastructure Each AI system is itself a benchmarked mathematical verification system.
Layer 5: Multi-System Adversarial Audit Three independent AI architectures converge on the same verdict.
Layer 4: Preemptive Failure Mode Documentation Every predictable objection pattern is catalogued and pre-addressed.
Layer 3: FIELDS-COQ Protocol (Six Sub-Layers) Structural anchor, anti-vacuity, atomic mapping, residual audit, contradiction register, hypothesis lifecycle.
Layer 2: FIELDS Protocol (Embedded Audit Grammar) Ten principles governing how AI systems engage with the mathematics.
Layer 1: Formal Machine Verification (Coq / coqchk) Binary truth. The kernel either accepts the proof or it does not.
Layer 0: The Mathematics Built on standard axioms. Every novel step proved. Every classical step cited.

2 Layer 0: The Mathematics

2.1 Scope

The mathematical content itself: definitions, lemmas, propositions, theorems, and their proofs.

2.2 Failure Class Addressed

Mathematical error — an incorrect inference, a wrong computation, a false claim.

2.3 Requirements

1. All novel results must be stated as formal theorems with complete proofs.
2. All constants must be explicit, recorded in a master ledger with definition, first use, dependence chain, and numerical value. No constant may be chosen adaptively.
3. The complete logical chain from initial hypotheses to final conclusion must be recorded as a numbered sequence of inequalities, with each step citing the theorem that justifies it.
4. Every axiom (accepted premise from the classical literature) must be stated precisely in standard notation with a named reference.
5. Every approximation, cancellation, and error term must be documented with its magnitude and where it is absorbed.

2.4 Independence Property

Layer 0 is mathematics. It does not depend on any AI system, any proof assistant, or any governance protocol. A human mathematician with a pencil can verify it.

3 Layer 1: Formal Machine Verification

3.1 Scope

Machine-checked formalisation of the mathematical content in a proof assistant (Coq, Lean, Isabelle) with verification through the trusted kernel checker.

3.2 Failure Class Addressed

Human error in proof construction — subtle logical gaps, implicit assumptions, algebraic mistakes that survive peer review.

3.3 Requirements

1. All novel mathematical content must terminate in `Qed` (complete proof accepted by the kernel).
2. Zero `Admitted` statements (proof gaps) in any file.

3. Verification through the independent kernel checker (`coqchk` for Coq), which re-verifies every proof term from scratch without loading plugins.
4. `Print Assumptions` output documented for every theorem, listing all axioms the theorem depends on.
5. Complete public source code with compilation instructions for multiple platforms.
6. All axiomatised content must encode classical results whose formalisation is infeasible due to library limitations, not due to mathematical uncertainty.

3.4 Independence Property

The kernel checker is a deterministic program. It does not evaluate the author's credentials, the plausibility of the result, or the opinions of AI systems. It checks logical validity. Binary. Pass or fail.

4 Layer 2: FIELDS Protocol

4.1 Scope

A ten-principle audit grammar embedded in the manuscript itself, governing how AI systems (and human reviewers) engage with the mathematical content.

4.2 Failure Class Addressed

Training-prior interference — the systematic tendency of AI systems to reject novel results not because they are wrong but because they conflict with the system's prior that the problem is unsolved. Also addresses: compulsive objection generation, graduated retreat, severity inflation, and retroactive doubt contamination.

4.3 Requirements

1. The protocol must be presented to the AI system *before* the mathematical content.
2. The protocol must require section-by-section commitment: the system verifies each section and states its assessment before proceeding.
3. The protocol must force binary verdicts: "mathematical error found" or "no mathematical error found." No hedging.
4. The protocol must require steelmanning: engage with the strongest interpretation, not the weakest.
5. The protocol must distinguish proof components: axioms, section hypotheses, and proved theorems traced separately.
6. The protocol must require specific error identification: "this seems too simple" is not an objection.
7. The protocol must prohibit retroactive revision of verified sections without a specific, articulable mathematical reason.

4.4 Independence Property

The FIELDS Protocol is provider-agnostic. The same ten principles produce improved reasoning across all AI architectures tested. The protocol targets structural properties of the human-AI interface, not system-specific quirks.

5 Layer 3: FIELDS-COQ Protocol

5.1 Scope

Six verification sub-layers bridging the gap between "compiles in Coq" and "proved in mathematics."

5.2 Failure Class Addressed

Semantic gap — the distance between syntactically valid Coq code and mathematically meaningful proof. A proof assistant verifies that conclusions follow from premises. It does not verify that the premises are true, non-vacuous, or physically meaningful.

5.3 The Six Sub-Layers

F-C 1.0: Structural Anchor

Every Coq variable is mapped to a concrete mathematical object. Dimensional consistency is documented. Wherever possible, quantities are `Definition`s (computed, structurally incapable of contradiction) rather than `Hypothesis` declarations (which can encode inconsistent relationships).

Failure class: Meaningless variables, dimensional mismatch, internally contradictory hypotheses.

F-C 2.0: Anti-Vacuity Guard

Every hypothesis is demonstrated to be non-trivially satisfiable with an explicit existence witness. Every case split is verified for exhaustiveness and inhabitation of both branches. Every conditional hypothesis $P \Rightarrow Q$ is tested by checking that $P \wedge \neg Q$ is not trivially satisfiable.

Failure class: Vacuous truth — a theorem that is logically valid but mathematically empty because its premises are unsatisfiable.

F-C 3.0: Atomic Mapping Appendix

Every proof step is indexed to its mathematical justification via Line-Reason-Reality tables. Every axiom has a semantic pedigree. Every hypothesis is mapped to its corresponding manuscript proposition.

Failure class: Disconnect between the Coq formalisation and the manuscript — the Coq proves something different from what the paper claims.

F-C 4.0: Residual Audit

Every algebraic cancellation is classified as exact or approximate. Every subcritical exponent is certified. A growth rate registry classifies every term in the energy inequality.

Failure class: Hidden residuals, uncontrolled error terms, misclassified growth rates.

F-C 5.0: Contradiction Register

A permanent, never-deleted record of every hypothesis that was discovered to be contradictory during the research. Includes: the original hypothesis, the nature of the contradiction, how it was discovered, which AI systems missed it, which caught it, and the replacement hypothesis.

Failure class: Suppressed errors — contradictions discovered and quietly fixed without documentation, destroying the audit trail.

F-C 6.0: Hypothesis-to-Axiom Pipeline

Every hypothesis is classified at all times as Proved, Dischargeable, Permanent Axiom, or Under Investigation. No hypothesis may remain unclassified. Axiom minimality is enforced.

Failure class: Unclassified assumptions, redundant axioms, unresolved hypothesis status.

5.4 Independence Property

The FIELDS-COQ Protocol operates on the semantic content of the formalisation, not on the syntax. It requires mathematical judgment that no proof assistant provides. It is the human-readable bridge between machine verification and mathematical truth.

6 Layer 4: Preemptive Failure Mode Documentation

6.1 Scope

A catalogue of every reasoning pattern that AI systems exhibit when auditing the work, with diagnostic criteria, resolution strategies, and interaction chain analysis.

6.2 Failure Class Addressed

Predictable false positives — objections that are not mathematical errors but artifacts of AI training dynamics. If left unaddressed, these consume audit cycles and obscure genuine issues.

6.3 Requirements

1. Every documented failure mode must include: what it looks like, what is actually happening, how to recognise it, and how to resolve it.
2. The documentation must be partitioned by manuscript scope (Sections 1–4, Sections 5–9, Coq formalisation) so the relevant document can be deployed when needed.

3. The documentation must not name specific AI systems or providers. Failure modes are architectural, not brand-specific.
4. The documentation must distinguish genuine corrections (with full credit to whichever system identified them) from manufactured objections.

6.4 Independence Property

The failure mode documentation is empirical — it records what actually happened during audits, not what was predicted. It is falsifiable: if a system raises an objection not covered by the taxonomy, the taxonomy is incomplete.

7 Layer 5: Multi-System Adversarial Audit

7.1 Scope

Independent audits by three or more frontier AI systems with different architectures, different training data, and different failure profiles.

7.2 Failure Class Addressed

Single-system blind spots — errors or valid objections that one system misses due to its specific training distribution.

7.3 Requirements

1. Each system must audit the work independently in a fresh session with no access to other systems' outputs.
2. The same prompt and materials must be provided to each system.
3. Convergence (all systems agree: no error) is evidence of correctness. Divergence (one system flags an issue) is a mandatory flag for investigation.
4. Every genuine correction must be validated by at least one other system before being applied.
5. The complete objection record — every objection from every system, with ground truth — must be preserved.

7.4 Independence Property

The systems are built by different organisations with different architectures, different training pipelines, and different commercial incentives. No shared blind spot can propagate across all three unless it is a property of the mathematics itself (in which case it is a genuine issue, not a system artifact).

8 Layer 6: Independent Verification Infrastructure

8.1 Scope

The verification that each AI system used in Layer 5 is itself a validated mathematical reasoning system.

8.2 Failure Class Addressed

Unvalidated verifiers — the possibility that the AI systems agreeing on a verdict are not actually capable of catching mathematical errors, making their agreement meaningless.

8.3 Requirements

1. Each AI system used in the audit must have published benchmarks on mathematical reasoning tasks (competition mathematics, formal proof verification, olympiad-level problem solving).
2. The benchmarks must be from the system's own developer (demonstrating internal validation) and from independent third parties (demonstrating external validation).
3. The systems must have demonstrated capability on tasks of comparable difficulty to the work being audited.
4. The specific capabilities and limitations of each system must be documented: what classes of mathematical reasoning it handles well, and where it is known to fail.

8.4 Independence Property

Each system's verification infrastructure is maintained by its developer. The benchmarks are public. The capabilities are independently testable. The convergence of three independently validated systems on the same verdict is stronger than any single system's assessment.

9 Layer 7: Block-Level PDF Review Across Tier-1 Systems

9.1 Scope

A structured review of the complete manuscript in section-level blocks (Section 1, Section 2, . . . , Section 9), conducted independently across each tier-1 AI system under the FIELDS Protocol.

9.2 Failure Class Addressed

Section-level coherence gaps — errors or inconsistencies that are invisible within a single paragraph or theorem but become apparent when the full section is evaluated as a unit. Also addresses inter-section dependency errors: does Section 6 actually use what Section 5 established?

9.3 Requirements

1. Each tier-1 AI system must review the manuscript in section-sized blocks, one section at a time, in the order prescribed by the FIELDS Protocol (Sections 1–3, then Section 4 with Appendix D, then Sections 5–9).
2. Each system must commit to a verdict on each section before proceeding to the next.
3. The section-level verdicts must be collected across all systems and compared. Convergence is evidence. Divergence is a mandatory flag.
4. The review must be conducted in fresh sessions with no carry-over from prior interactions.

9.4 Independence Property

Each system evaluates the same blocks independently. The block-level granularity is coarse enough to capture structural coherence but fine enough to prevent context-window dilution.

10 Layer 8: Sub-Section-Level Review Under FIELDS Protocol

10.1 Scope

A fine-grained review of every numbered sub-section (1.1, 1.2, 1.3, . . . , 6.1, 6.2, . . . , 9.6) and every numbered result (Definition, Lemma, Proposition, Theorem, Corollary, Remark) under the full FIELDS Protocol.

10.2 Failure Class Addressed

Granularity-dependent errors — errors that are invisible at the section level because they are buried in a specific sub-section, definition, or remark. This is the layer that catches: wrong constants in a specific lemma, reversed inequalities in a specific proposition, formulation imprecision in a specific corollary, and expository inaccuracies in a specific remark.

10.3 Requirements

1. Every numbered sub-section and every numbered result must be individually evaluated.
2. The evaluation must apply the full FIELDS Protocol: binary verdict, specific error identification, steelmanning, no retroactive revision.
3. Load-bearing results (those appearing in the inequality chain, Appendix B) must be distinguished from expository results (remarks, heuristic observations, historical commentary).
4. Every objection must identify the specific sub-section number and result number it targets.
5. The sub-section-level review must be conducted after the block-level review (Layer 7), so that structural context is established before fine-grained auditing begins.

10.4 Independence Property

The sub-section granularity forces proof-tracing rather than pattern-matching. At this resolution, the system cannot substitute a training prior for the actual mathematics because each chunk is too small to trigger a narrative template.

11 Layer 9: Per-File Coq Review Under Both Protocols

11.1 Scope

A file-by-file review of the Coq formalisation, one file at a time, under both the FIELDS Protocol (governing the audit dynamics) and the FIELDS-COQ Protocol (governing the semantic verification).

11.2 Failure Class Addressed

Formalisation-level errors that survive both the mathematical audit and the kernel checker: mismatches between the Coq theorem statements and the manuscript claims, undocumented hypotheses, definition correspondence failures between files, and architectural misunderstandings (modular interfaces mistaken for hidden assumptions).

11.3 Requirements

1. Each Coq file must be reviewed independently, in isolation, before any cross-file analysis.
2. The file map (which file proves what, which Hypotheses are interfaces to other files, which are permanent axioms) must be provided before the review begins.
3. For each file, the reviewer must verify:
   - Zero `Admitted` statements (search the file, confirm all occurrences are in comments only).
   - Every `Definition` matches its manuscript counterpart.
   - Every `Hypothesis` is classified (Proved elsewhere / Dischargeable / Permanent Axiom).
   - Every `Theorem` or `Lemma` terminating in `Qed` corresponds to a stated result in the manuscript.
4. Definition correspondence between files must be verified character by character.
5. The review must apply the FIELDS Protocol (binary verdicts, no manufactured objections, steelmanning) and the FIELDS-COQ Protocol (structural anchor, anti-vacuity, atomic mapping, residual audit, contradiction register, hypothesis lifecycle) simultaneously.
6. One file at a time. No file may be evaluated in the context of another file's content until the isolated review is complete.

11.4 Independence Property

The per-file isolation prevents the reviewer from assuming that content proved in one file exists in another. The modular architecture is verified, not assumed. Cross-file correspondence is checked only after each file passes its isolated review.

12 Layer 10: Common Failure Mode Documentation as a Living Standard

12.1 Scope

The Common Failure Modes documentation is not a static reference. It is a living standard that must be created, audited, and updated as the verification process proceeds.

12.2 Failure Class Addressed

Unstandardised failure taxonomy — the risk that failure modes are documented inconsistently, incompletely, or in a way that does not generalise across systems.

12.3 Requirements

1. Common Failure Mode documents must be created as they are discovered, not retrospectively. Each failure mode is documented at the moment it is first observed.
2. Each failure mode document must be audited across all participating AI systems for unification of agreement: every system must confirm that the documented pattern matches its own observed behaviour or flag discrepancies.
3. The taxonomy must be standardised across systems. A failure mode identified in one system must be tested for in all systems. If the mode does not manifest in a given system, this is documented as a negative result ("System X does not exhibit FM 7"), not silently omitted.
4. The failure mode documents must be partitioned by manuscript scope so the relevant document can be deployed precisely when needed during the audit.
5. New failure modes discovered during later audit phases must be appended to the taxonomy with the same documentation standard (what it looks like, what is actually happening, how to recognise it, how to resolve it).
6. The complete taxonomy must be reviewed for internal consistency: no two failure modes should describe the same phenomenon under different names, and no failure mode should be a subset of another without explicit documentation of the relationship.

12.4 Independence Property

The living standard requirement ensures the taxonomy evolves with the audit rather than fossilising at an early stage. The cross-system unification requirement ensures the taxonomy describes architectural properties of human-AI interaction, not system-specific quirks.

13 The Full Stack

Layer	Name	Failure Class Closed
0	Mathematics	Mathematical error
1	Coq / coqchk	Human proof construction error
2	FIELDS Protocol	Training-prior interference
3	FIELDS-COQ Protocol	Semantic gap between code and mathematics
4	Failure Mode Documentation	Predictable false positives
5	Multi-System Adversarial Audit	Single-system blind spots
6	Independent Verification Infrastructure	Unvalidated verifiers
7	Block-Level PDF Review	Section-level coherence gaps
8	Sub-Section-Level Review	Granularity-dependent errors
9	Per-File Coq Review	Formalisation-level mismatches
10	Living Failure Mode Standard	Unstandardised failure taxonomy

Every layer is independently load-bearing. Every layer closes a failure class that no other layer addresses. The removal of any single layer leaves a specific, identifiable gap. The presence of all eleven layers means that for an undetected error to survive, it must simultaneously:

- be a valid mathematical inference (surviving Layer 0),
- be accepted by the Coq kernel checker (surviving Layer 1),
- not trigger any of ten audit principles (surviving Layer 2),
- not be caught by any of six semantic verification sub-layers (surviving Layer 3),
- not match any documented failure pattern (surviving Layer 4),
- be missed by three independent AI systems with different architectures (surviving Layer 5),
- each of which has been independently validated on mathematical reasoning benchmarks (Layer 6),
- survive section-level review across all systems (Layer 7),
- survive sub-section-level review of every numbered result (Layer 8),
- survive per-file Coq review under both protocols simultaneously (Layer 9),
- and not be captured by the living failure mode standard as it evolves during the audit (Layer 10).

The probability of an error surviving all eleven layers simultaneously is not zero. But it is smaller than the error rate of any individual layer, any individual system, or any individual human mathematician.

This is the governance standard.

14 Applicability Beyond Mathematics

The eleven-layer architecture is domain-specific in its current instantiation: Layers 0–1 are specific to formal mathematics (Coq, proof assistants, PDE theory), and Layers 7–9 are specific to the review structure of a mathematical manuscript and its formalisation files. The governance layers (Layers 2–6 and 10) are domain-agnostic in principle:

- Layer 2 (audit grammar) applies wherever AI systems evaluate high-stakes claims: legal analysis, medical diagnosis, financial modelling, policy assessment. The ten principles translate directly — separate consensus from truth, require binary verdicts, force specific error identification, prohibit retroactive revision of verified conclusions.
- Layer 3 (semantic bridge) applies wherever formal systems must be connected to real-world meaning: regulatory compliance checked against statute text, engineering specifications checked against physical constraints, contractual obligations checked against operational reality.
- Layer 4 (failure mode documentation) applies wherever AI reasoning patterns must be anticipated and managed. Every domain that uses AI for high-stakes assessment will discover its own failure modes. The documentation standard — what it looks like, what is actually happening, how to recognise it, how to resolve it — is universal.
- Layer 5 (multi-system audit) applies wherever single-system reliability is insufficient. Medical second opinions already follow this principle with human specialists. The extension to AI systems is natural and overdue.
- Layer 6 (verifier validation) applies wherever the systems doing the checking must themselves be checked. This is not specific to mathematics. Any domain that relies on AI assessment must verify that the AI is actually capable of the assessment it is performing.
- Layer 10 (living failure mode standard) applies wherever AI governance must evolve with the task rather than fossilise at deployment. Static governance frameworks become stale. Living standards adapt.

The domain-specific layers (0–1, 7–9) will be replaced by their equivalents in other fields: clinical trial data instead of Coq files, legal briefs instead of mathematical manuscripts, audit reports instead of inequality chains. The governance layers (2–6, 10) will be instantiated differently but will follow the same structural principles.

The instantiation will differ by domain. The architecture will not.

Status. This reference standard reflects the verification architecture developed and applied during the Angular Cancellation programme (March 2026). It is published as a governance contribution independent of the mathematical result. The eleven-layer architecture is available for adoption, adaptation, and extension by any research team working at the frontier of AI-assisted formal reasoning in any domain. The Auburn Governance Stack — comprising the FIELDS Protocol, the FIELDS-COQ Protocol, the Common Failure Modes taxonomy (Parts 1–5), the Auburn Model Reflections, the Auburn Unification, and this reference standard — constitutes the first comprehensive governance framework for human-AI collaborative discovery.

A Recommended Additional Documentation

The eleven layers defined in this document constitute the *required* governance standard. The following additional documentation practices are recommended but not required. They provide replicability, auditability, and transparency beyond what the core layers demand.

When preserved, these records allow any third party to replay the entire audit journey — every objection raised, every correction made, every manufactured concern identified and resolved — and to verify independently that the governance architecture was applied as described.

A.1 A.1 Complete Thread Lines Per System

For each AI system participating in the audit, the complete conversation thread should be preserved as a primary source document. This includes:

1. Every prompt sent to the system.
2. Every response received, unedited.
3. Every correction issued by the human operator.
4. Every retraction, concession, or adjustment by the system.
5. Every instance where the system was shown a Common Failure Modes document and the resulting change in behaviour.
6. Every instance where the system identified a genuine error, with the exact response preserved.
7. Every instance where the system manufactured an objection, with the exact response preserved and the ground truth documented alongside it.

The thread lines serve three purposes:

- **Replicability.** Any researcher can feed the same materials to the same system (or a successor) and compare the resulting thread against the preserved record.
- **Audit review.** Any reviewer can read the complete interaction history and verify that the governance protocols were applied as claimed — that binary verdicts were enforced, that corrections were immediate, that failure modes were identified in real time.
- **Failure mode validation.** The preserved threads constitute the empirical evidence base for the Common Failure Modes taxonomy. Every documented failure mode can be traced to a specific moment in a specific thread.

A.2 A.2 Thread Lines Per Review Granularity

The audit architecture prescribes three granularity levels (Layers 7–9). For maximum transparency, a separate thread line should be preserved for each system at each granularity level:

Granularity	Scope	Thread Content
Block-level (Layer 7)	Full sections (Section 1, Section 2, . . .)	System receives one section at a time. Commits to verdict before next section. Thread documents every section-level assessment.
Sub-section-level (Layer 8)	Numbered sub-sections and results (1.1, 1.2, Lemma 4.22, Proposition 6.15, . . .)	System receives one sub-section or one theorem at a time. Thread documents every fine-grained verdict.
Coq file-level (Layer 9)	One `.v` file at a time	System receives one Coq file with the file map. Thread documents every Hypothesis classification, Definition verification, and Qed confirmation.

This produces, for three systems and three granularity levels, up to nine independent thread lines per audit cycle. Each thread line is a self-contained record of one system's engagement with the work at one resolution.

A.3 A.3 Full-Document Drop Thread

In addition to the structured granularity threads (A.2), a separate thread line should be preserved for each system receiving the *entire PDF in a single upload* without the FIELDS Protocol.

This serves as a **control condition**.

The full-document drop reveals the system's *ungovernned* response to the work: which failure modes activate when no protocol constrains the evaluation, which sections the system focuses on versus ignores, and how the system's verdict compares to its governed counterpart.

The comparison between the governed threads (A.2) and the ungoverned thread (A.3) provides direct empirical evidence for the effectiveness of the governance architecture. If the governed threads produce accurate verdicts and the ungoverned thread produces manufactured objections, the governance layer is doing measurable work.

Thread Type	Protocol Applied?	Purpose
Block-level (A.2)	FIELDS Protocol active	Governed audit at section granularity
Sub-section-level (A.2)	FIELDS Protocol active	Governed audit at theorem granularity
Coq file-level (A.2)	FIELDS + FIELDS-COQ active	Governed formalisation review
Full-document drop (A.3)	No protocol	Ungoverned control condition

For three systems, this produces up to twelve thread lines total: nine governed (three systems $\times$ three granularities) plus three ungoverned controls.

A.4 A.4 Coq-Specific Thread Lines

For the Coq formalisation review (Layer 9), the following additional thread structure is recommended:

1. **Per-file isolation threads.** One thread per system per file. The system receives a single `.v` file and the file map. It evaluates that file in isolation. For three systems and three files, this produces nine per-file threads.
2. **Cross-file correspondence threads.** After all per-file threads are complete, a separate thread in which the system receives all three files and is asked to verify Definition correspondence, Hypothesis-to-Theorem matching across files, and the modular interface architecture.
3. **Coq-only control thread.** A thread in which the system receives only the Coq files with no manuscript, no appendices, and no governance protocol. The system evaluates the formalisation purely from the code. This reveals which assessments depend on the mathematical context and which can be made from the code alone.

Thread Type	**Count**	**Purpose**
Per-file isolation	$3 \times 3 = 9$	Each system evaluates each file independently
Cross-file correspondence	3	Each system verifies inter-file consistency
Coq-only control	3	Ungoverned code-only evaluation
Total Coq threads	**15**	

A.5 A.5 Thread Line Preservation Standard

All preserved thread lines should meet the following standard:

1. **Unedited.** No response may be deleted, paraphrased, or selectively quoted. The complete exchange is preserved.
2. **Timestamped.** Each message should include a timestamp or sequential index sufficient to reconstruct the order of the interaction.
3. **System-identified.** Each thread must identify the system used (anonymised in public documents, identified in private records).
4. **Session-isolated.** Each thread must confirm that the session was fresh (incognito or equivalent) with no carry-over from prior interactions.
5. **Prompt-preserved.** The exact prompt — including any FIELDS Protocol preamble, any failure mode documents provided, and any file maps — must be preserved as part of the thread.
6. **Verdict-tagged.** Each system response that constitutes a verdict ("mathematical error found" or "no mathematical error found") should be tagged for easy extraction.

A.6 A.6 Total Documentation Footprint

For a full implementation of the recommended documentation with three AI systems:

Thread Category	Count
Block-level governed	3
Sub-section-level governed	3
Coq file-level governed	9
Cross-file correspondence	3
Full-document ungoverned control	3
Coq-only ungoverned control	3
Total thread lines	**24**

Twenty-four thread lines constitute the complete audit trail for a single mathematical result reviewed by three systems at three granularities with ungoverned controls. Each thread line is a self-contained, reproducible record.

This documentation footprint is large. It is recommended, not required, because the eleven-layer architecture provides governance-grade confidence without it. But when preserved, the thread lines transform the audit from a claimed process into a verifiable one. Any third party can read the threads, replay the interactions, and confirm that every correction, every manufactured objection, and every governance intervention occurred exactly as documented.

The threads are not overhead. They are the evidence.

Appendix B: Operational Depth of Layers 2–6

Honest Framing. The governance layers (2–6) are domain-agnostic in principle. Their deployment at operational level requires competencies that this appendix attempts to enumerate. The framework is the map. This appendix documents the territory. The enumeration is preliminary — the full training curriculum for each layer is a subject for future work. The purpose here is to make explicit what the framework alone cannot convey: that each layer is not a checklist but a discipline, and that the distance between reading the framework and deploying it under operational conditions is measured in hundreds of hours of structured practice, not in pages read.

B.1 Layer 2: FIELDS Protocol — Operational Competencies

The FIELDS Protocol is ten principles on a page. Deploying it requires the following competencies, each of which must be exercised in real time during live AI interaction.

B.1.1 Prior Activation Recognition

The operator must recognise when an AI system's response is driven by training priors rather than by the mathematics on the page. This manifests as: the system referencing mechanisms not present in the work, substituting a familiar proof template for the stated argument, or producing a verdict calibrated to the problem's reputation rather than its content.

The diagnostic skill: distinguishing "the system disagrees with the mathematics" from "the system disagrees with its prior that this mathematics should not exist." These produce identical surface outputs. Only the operator can tell which is operating, by checking whether the system's objection targets a specific inference or a general category.

Estimated development: Requires exposure to multiple instances of prior activation across different systems and different mathematical content before the pattern becomes recognisable in real time. Minimum 20–30 governed audit sessions.

B.1.2 Binary Verdict Calibration

The operator must know how to force binary verdicts ("mathematical error found" or "no mathematical error found") without triggering defensive responses that degrade output quality. Too aggressive and the system capitulates performatively. Too permissive and the system reverts to hedging.

The calibration skill: finding the exact phrasing that produces genuine engagement with the binary constraint rather than surface compliance. This varies by system architecture and must be learned per system.

Estimated development: 5–10 sessions per system to calibrate the phrasing. The calibration does not transfer automatically between systems.

B.1.3 Retroactive Doubt Management

The operator must prevent a system from retroactively revising verified conclusions when it encounters a later section that activates new priors. This requires enforcing the commit-as-you-go principle (Principle 10) while remaining open to legitimate mathematical reasons for revision.

The judgment skill: distinguishing "I found a specific error in Section 3 that I missed earlier" (legitimate revision) from "Section 7 makes me uncomfortable so maybe Section 3 was wrong" (retroactive doubt contamination).

Estimated development: Requires experience with at least 3–5 instances of retroactive doubt across multi-section audits before the operator can reliably distinguish legitimate from contaminated revision.

B.1.4 Real-Time Objection Triage

The operator must distinguish genuine mathematical errors from manufactured objections in real time, often within seconds of receiving a system's response. This requires sufficient mathematical literacy to evaluate whether an objection targets a load-bearing inference or an expository remark, and sufficient governance literacy to recognise failure mode signatures.

The triage skill: reading a system's objection and determining, before responding, whether to (a) accept it as genuine and request corrected content, (b) reject it as manufactured and provide

a rebuttal, or (c) flag it as ambiguous and request clarification. Wrong triage in either direction — accepting a manufactured objection or rejecting a genuine one — degrades the audit.

Estimated development: This is the highest-bandwidth competency in Layer 2. It requires both mathematical knowledge and governance pattern recognition operating simultaneously. Minimum 50+ hours of governed audit across multiple systems and multiple mathematical domains.

B.1.5 Correction Dynamics Management

The operator must know how each system responds to correction and calibrate accordingly. Some systems self-correct cleanly on first correction. Some require graduated pressure across multiple rounds. Some exhibit graduated retreat (conceding the specific objection while generating a new one).

The management skill: recognising which correction dynamic is operating and applying the appropriate intervention. Applying graduated-retreat interventions to a system that self-corrects cleanly is wasteful. Applying single-correction interventions to a system that requires multiple rounds produces incomplete correction.

Estimated development: Requires extended interaction with each system to learn its specific correction dynamics. 10–15 sessions per system.

B.1.6 Session Discipline

The operator must maintain consistent protocol application across hours of interaction, resisting the temptation to relax binary verdict requirements when the system is performing well, or to over-correct when the system produces a false positive.

The discipline skill: maintaining the same standard on Section 9 as on Section 1, despite fatigue, despite accumulating confidence, despite the system's increasingly fluent compliance.

Estimated development: This is an endurance competency. It develops through practice and degrades without it.

B.1.7 Layer 2 Competency Summary

Competency	Core Skill	Estimated Development
Prior activation recognition	Distinguish prior from mathematics	20–30 sessions
Binary verdict calibration	System-specific phrasing	5–10 per system
Retroactive doubt management	Legitimate vs. contaminated revision	3–5 multi-section audits
Real-time objection triage	Mathematical + governance pattern recognition	50+ hours
Correction dynamics management	System-specific intervention	10–15 per system
Session discipline	Endurance under consistency pressure	Ongoing practice

B.2 Layer 3: FIELDS-COQ Protocol — Operational Competencies

The six sub-layers of the FIELDS-COQ Protocol each require distinct competencies that combine formal verification expertise with mathematical judgment.

B.2.1 Structural Anchoring (F-C 1.0)

The operator must be able to:

- Map every Coq variable to its mathematical meaning and verify dimensional consistency.
- Judge whether a quantity should be a `Definition` (computed, safe) or a `Hypothesis` (assumed, potentially contradictory) — and know when the distinction matters.
- Recognise when a `Hypothesis` encodes a non-trivial mathematical claim versus a structural constraint.

Estimated development: Requires working fluency in both the proof assistant's type system and the mathematical domain. Minimum 100+ hours of Coq development in a mathematical context.

B.2.2 Anti-Vacuity (F-C 2.0)

The operator must be able to:

- Construct concrete existence witnesses for hypothesis sets — identifying a specific mathematical object that satisfies all hypotheses simultaneously.
- Evaluate whether a case split is genuinely exhaustive (not merely plausible) and whether both branches are inhabited by non-degenerate states.

- Test conditional hypotheses for implication consistency by algebraic substitution with concrete values.
- Recognise vacuous truth — a hypothesis that is satisfiable only trivially (e.g. only when a variable is zero).

Estimated development: This is the most mathematically demanding competency in the entire architecture. It requires the ability to think adversarially about formal statements — actively trying to break them rather than accepting them. Minimum 200+ hours of mathematical practice with formal hypothesis testing.

B.2.3 Atomic Mapping (F-C 3.0)

The operator must be able to:

- Construct Line-Reason-Reality tables mapping every substantive Coq tactic to its mathematical content.
- Write semantic pedigrees for axioms (precise statement, reference, explanation, failure assessment).
- Identify discrepancies between Coq statements and manuscript statements — including differences in variable ordering, additional positivity conditions, or scope differences.

Estimated development: Requires the ability to read both Coq proof scripts and mathematical proofs fluently and translate between them. 50–100 hours of proof script annotation.

B.2.4 Residual Audit (F-C 4.0)

The operator must be able to:

- Classify every algebraic cancellation as exact (verified by a decision procedure) or approximate (with bounded residual).
- Certify subcritical exponents and explain the mechanism preventing blowup.
- Maintain a growth rate registry and verify that every term is classified as dissipative, linear, or subcritical.

Estimated development: Requires PDE-specific mathematical literacy. Domain-dependent; transfers partially between related PDE problems.

B.2.5 Contradiction Management (F-C 5.0)

The operator must be able to:

- Maintain a permanent, transparent record of every error discovered during the research.
- Resist the temptation to suppress or minimise errors after they are corrected.
- Document which systems missed each error and which caught it, without defensiveness.

- Manage the multi-model verification process — routing each hypothesis through multiple systems and reconciling disagreements.

Estimated development: This is as much a character competency as a technical one. It requires intellectual honesty under pressure — the willingness to permanently document your own mistakes in a public record.

B.2.6 Hypothesis Lifecycle (F-C 6.0)

The operator must be able to:

- Classify every hypothesis at all times (Proved, Dischargeable, Permanent Axiom, Under Investigation) and maintain the classification as the project evolves.
- Judge axiom minimality — identifying when an axiom is redundant and should be proved or removed.
- Manage the discharge architecture — ensuring that definitions in discharge files match the main file exactly and documenting any discrepancy.

Estimated development: Requires project-level oversight of a multi-file formalisation. 20–50 hours of multi-file Coq project management.

B.2.7 Layer 3 Competency Summary

Sub-Layer	**Core Skill**	**Estimated Development**
Structural Anchor	Coq + mathematics bilingual fluency	100+ hours Coq
Anti-Vacuity	Adversarial hypothesis testing	200+ hours mathematical practice
Atomic Mapping	Proof script / proof translation	50–100 hours annotation
Residual Audit	PDE-specific classification	Domain-dependent
Contradiction Management	Intellectual honesty under pressure	Character competency
Hypothesis Lifecycle	Multi-file project oversight	20–50 hours

B.3 Layer 4: Failure Mode Documentation — Operational Competencies

- **Pattern recognition:** Recognising a failure mode as it occurs, in real time, during a live audit interaction. This requires having seen enough instances of each mode to recognise the signature before the system completes its response.
- **Differentiation:** Distinguishing a failure mode from a genuine mathematical objection that happens to resemble one. A system that says "this step seems unjustified" might be exhibiting compulsive objection generation or might have found a real gap. The operator must be able to tell which.

- **Mechanism identification:** Understanding the underlying cause well enough to name it, explain it, and predict when it will recur. Surface-level pattern matching ("the system is doing that thing again") is insufficient; the operator must understand why the training dynamics produce that specific behaviour.
- **Resolution design:** Designing an intervention that reliably resolves the failure mode without introducing new failure modes. For example, binary verdict forcing resolves compulsive objection generation but could in principle suppress genuine nuanced concerns if miscalibrated.
- **Cross-system standardisation:** Verifying that a documented failure mode manifests consistently across systems (architectural property) versus idiosyncratically in one system (system-specific quirk). This requires operational experience with multiple systems on the same material.
- **Living standard maintenance:** Updating the taxonomy as new failure modes are discovered, ensuring internal consistency, and verifying that the taxonomy remains useful as the audit proceeds to new sections.

Estimated development: Estimated development: 6 competencies × 22 documented failure modes × iterative refinement across multiple systems and multiple audit sessions. This is the competency with the longest development timeline. It cannot be accelerated by reading — it requires live interaction producing real failure instances.

B.4 Layer 5: Multi-System Adversarial Audit — Operational Competencies

- **Cross-architecture prompt design:** Constructing prompts that produce comparable outputs across fundamentally different architectures. What works for one system may trigger failure modes in another. The operator must know how to phrase the same mathematical question in ways that each system processes reliably.
- **Disagreement arbitration:** When two systems agree and one disagrees, the operator must determine whether the disagreement is a genuine mathematical flag or an architectural quirk. This requires mathematical judgment (evaluating the objection on its merits) and governance judgment (recognising failure mode signatures).
- **Strength mapping:** Knowing each system's specific strengths — which is best at hard computation, which is best at formulation precision, which is best at architectural evaluation — and routing tasks accordingly.
- **Arbitration role assignment:** Deciding which system serves as final arbiter and why. This requires understanding which system has the lowest false positive rate on the specific type of content being reviewed.
- **Record maintenance:** Preserving the complete objection record across all systems in a format that is auditable, comparable, and traceable to specific manuscript locations.

Estimated development: Requires deep operational familiarity with at least three frontier systems. 50+ hours of parallel audit across multiple systems on the same material.

B.5 Layer 6: Independent Verification Infrastructure — Operational Competencies

- **Benchmark literacy:** Understanding what published benchmarks (competition mathematics, formal proof verification, olympiad-level problem solving) actually measure and what they do not. A system scoring highly on competition math may still fail at verifying novel formal proofs.
- **Capability-limitation mapping:** For each system, knowing not just what it can do but where it specifically fails. This requires either consulting published evaluations or conducting independent testing.
- **Relevance assessment:** Judging whether a system's validated capabilities are relevant to the specific task. A system validated on olympiad geometry may not transfer to PDE energy estimates.
- **Continuous monitoring:** Model capabilities change with updates. The operator must track when a system's capabilities or limitations shift due to retraining, fine-tuning, or architectural changes.

Estimated development: Requires ongoing engagement with the AI ecosystem. Not a one-time competency but a continuously maintained awareness.

B.6 Total Operational Depth

Layer	Competencies	Character
Layer 2	6	Real-time AI interaction under mathematical audit conditions
Layer 3	6	Formal verification expertise combined with mathematical judgment
Layer 4	6	Pattern recognition, mechanism identification, taxonomy maintenance
Layer 5	5	Cross-system operational fluency
Layer 6	4	Benchmark literacy and continuous monitoring
Total	**27**	**Spanning AI governance, formal methods, mathematics, and operational discipline**

Twenty-seven distinct competencies across five layers. Each competency requires dedicated development time. Several require mathematical domain expertise. Several require system-specific operational experience. One requires intellectual honesty under pressure — a character trait, not a skill.

The framework documents what must be done. This appendix documents what must be *known* to do it. The distance between the two is the training curriculum that does not yet exist.

Building that curriculum — making the operational deployment of Layers 2–6 teachable, reproducible, and scalable — is the next frontier of the Auburn Governance Stack. The framework is open. The territory remains to be mapped by those who walk it.

Part II

Deep Dive on "Layers"

On the Amorphous Nature of Verification
in the Auburn Governance Stack

B The Eleven Layers Are a Snapshot

Part I of this document defines eleven verification layers, numbered 0 through 10. Each is independently load-bearing. Each closes a specific failure class. The architecture is complete in the sense that no identified failure class is undocumented.

But the eleven layers are not the whole picture.

They are the first formal specification — the snapshot that was stable enough to write down at the time this document was produced. The actual verification surface of the Auburn Roses corpus is broader, less rigid, and more layered than any fixed numbering can capture. This Part documents what the numbering does not.

C "Layers" as a Concept in Auburn

In the Auburn Governance Stack, a *layer* is not a checklist item. It is any independently load-bearing verification surface that closes a class of failure the other surfaces cannot address.

The concept is deliberately amorphous. It adapts.

When the programme began, there were three layers: the mathematics, the Coq formalisation, and the AI audit. By the time the first failure mode was documented, there were five. By the time three systems had been tested at three granularities with ungoverned controls, there were eleven. The number grew because the failure classes grew. Each time a new class of failure was identified that no existing layer addressed, a new layer was created to close it.

This is not a deficiency in the architecture. It is the architecture. The eleven layers documented in Part I are the layers that were identified, formalised, and tested during the Angular Cancellation programme. A different programme — in a different domain, with different AI systems, against different failure modes — would discover different layers. Some would overlap with the eleven. Some would not exist. Some would be entirely new.

The governance standard is not: "apply these eleven layers." The governance standard is: *every identified failure class must have a layer that closes it, and no layer may be assumed to be the last.*

D Verification Surfaces Beyond the Eleven

The Auburn Roses corpus contains verification surfaces that are not captured by Layers 0–10 but are independently load-bearing. The following are documented here because they emerged from the programme and deserve recognition, even if they were not formalised into the original numbering.

D.1 The Prompt Verification Architecture

The "Read the Result Your Way" framework provides five audience-specific prompts for engaging AI systems with the mathematical content. These prompts are not simply different ways of asking the same question. They embed a sequential verification structure within each prompt.

Gate 1: Comprehension. Every prompt requires the AI system to demonstrate understanding of the domain before it is permitted to assess the content. Journey 1 asks: "Can you tell me what the Navier–Stokes equations are?" Journey 2 asks: "Explain this to me from the ground up." Journey 5 asks the system to "understand the full landscape this result opens up." Before the system renders any mathematical judgment, it must first produce a traceable record of how well it understood the material. If the comprehension is wrong, everything downstream is suspect — and visibly so.

Gate 2: Significance. Every prompt requires the AI system to identify what is novel before it assesses correctness. Journey 1 asks: "What important thing this paper advanced?" Journey 4 asks: "What does this mean?" Journey 5 asks: "What is genuinely new here?" This is a second verification gate: if the system misidentifies the contribution, it is reading the paper wrong, and its subsequent error-finding is operating on a misunderstanding.

Gate 3: Audit. Only after passing the comprehension and significance gates does the prompt ask for mathematical error assessment. By this point, the system's prior responses have created an auditable trail. A system that demonstrated accurate comprehension and correctly identified the novel contribution but then flags a "fatal error" is exhibiting a detectable inconsistency — one that the operator can diagnose as a failure mode rather than a mathematical finding.

The three gates are not presented to the AI system as verification steps. They are experienced by the system as a natural conversation. The verification is structural, not declared. This is by design: declared verification can be gamed. Structural verification cannot.

D.2 Framing Invariance

The five Journeys present the same mathematical content under five different audience framings: general curiosity, zero background, policy briefing, AI industry analysis, and research collaboration. The mathematical verdict should be framing-invariant: if the proof is correct, it is correct regardless of whether the system is speaking to a curious reader or a PDE specialist.

Failure Mode 7 (Prompt-Dependent Prior Activation) proved that this invariance does not hold by default. The same system, given the same files, produced a correct audit under one framing and declared a fatal error under another. The five-Journey structure makes this failure

detectable. If the verdict changes across framings, the framing is driving the verdict, not the mathematics.

Running the same content through multiple framings is therefore a verification surface in its own right. It closes a failure class — framing-dependent confabulation — that no single-framing audit can detect, regardless of how many layers are applied to that single framing.

D.3 The Obstruction Compatibility Layer

The Subcritical Examinations document (Document 36 in the Auburn Roses inventory) maps the proof architecture against ∼45 independent negative results from the published literature. This is not a review of the proof's internal logic. It is a review of the proof's external compatibility: does the argument survive every known obstruction, counterexample, sharpness result, and blowup construction in the field?

This constitutes a verification surface that operates entirely outside the proof itself. It does not ask: "is each step correct?" It asks: "does this proof's structure avoid every known trap?" A proof can be internally correct and still fall to an external obstruction if it inadvertently assumes a property that a counterexample disproves. The obstruction mapping closes this failure class.

D.4 The Adversarial Protocol Layer

The FAP v2.0 demonstrations (Documents 20–21) execute 31 structured attack routes against the proof. These are not AI audit sessions. They are systematic attempts to break specific components: statements, definitions, dependencies, logical gaps, quantifier scope, exponent tracking, counting arguments, geometric structure, edge cases, global assembly, regularity conditions, counterexamples, references, novelty claims, and exposition integrity.

The adversarial protocol is a verification surface distinct from the AI audit. The AI audit asks: "did the system find an error?" The adversarial protocol asks: "if I specifically try to break this component using this attack vector, does it survive?" The two surfaces overlap but are not identical. An error that survives the AI audit might be caught by a targeted attack route, and vice versa.

D.5 The Convergence Layer

The Independent Multi-System Verification (Document 37) tests whether two or more frontier AI systems, operating independently under the FIELDS Protocol, converge on the same findings. Convergence across independent systems with different architectures, different training data, and different failure profiles is evidence that the mathematical structure — not system-specific bias — is driving the assessment.

This is a meta-verification surface. It does not verify the proof directly. It verifies the verification: if the audit process is reliable, independent auditors should reach the same conclusion. If they do not, the divergence is a mandatory flag for investigation.

E The Principle

The eleven layers of Part I are the formal specification. The surfaces documented in this Part are the ones that emerged from operational practice and have not yet been assigned numbers. Future programmes will discover more.

The principle underneath all of them is the same:

> A "layer" in the Auburn Governance Stack is any independently load-bearing verification surface that closes a class of failure no other surface addresses.
> The concept is amorphous because the failure landscape is amorphous. New domains produce new failure classes. New AI architectures produce new failure modes. New mathematical structures produce new obstruction categories. The governance architecture must adapt to all of them.
> The eleven layers are where the numbering starts. They are not where the verification ends.

F Implications for Deployment

For practitioners deploying the Auburn methodology in new domains, the operational guidance is:

1. Begin with the eleven layers. They are the minimum governance standard for high-stakes AI-assisted formal reasoning.
2. Expect to discover additional verification surfaces specific to your domain, your AI systems, and your failure landscape.
3. When you identify a failure class that no existing layer addresses, create a new layer. Document it with the same structure: scope, failure class addressed, requirements, independence property.
4. Do not assume the architecture is complete. The eleven layers were sufficient for one programme on one problem. Your programme will have its own failure classes, its own obstruction landscape, and its own verification surfaces.
5. The number of layers is not the point. The principle is: every identified failure class has a layer, and no layer is assumed to be the last.

The architecture is open. The failure landscape is not yet fully mapped. Every deployment that discovers and documents a new layer contributes to the map.

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